# Interplanetary AI Nervous System: A Distributed Framework for Autonomous Space Communication and Decision-Making

Gopichand Kanaparthi
JNTUH, India
Email: Kanaparthigopichand265@gmail.com

***Abstract*—The rapid expansion of space missions demands a scalable and autonomous framework for communication and decision-making across interplanetary distances. This paper proposes the Interplanetary AI Nervous System (IANS), a distributed, intelligent architecture designed to function as the "neural backbone" for satellite constellations, planetary rovers, and deep-space probes. Inspired by biological nervous systems, IANS integrates artificial intelligence, distributed computing, and satellite-based communication to enable self-organization, adaptive routing, and autonomous fault recovery without constant human intervention. By leveraging deep learning algorithms, reinforcement learning, and real-time optimization, the system ensures efficient inter-satellite data flow, low-latency decision-making, and resilience to disruptions caused by harsh space environments. Experimental simulations demonstrate how IANS can outperform traditional centralized models in terms of reliability, adaptability, and energy efficiency.**

***Index Terms*—Interplanetary AI, Federated Learning, Delay-Tolerant Networks, Graph Neural Networks, Space Robotics, Autonomous Systems**

## I. Introduction

Interplanetary missions are rapidly growing in complexity: multiple rovers, orbiters, landers, and human habitats will coexist and cooperate in the coming decade. However, the physics of space imposes constraints (long round-trip times, intermittent connectivity, narrowband links) that make traditional centralized control impractical for timely autonomy. For example, Earth–Mars round-trip delays can approach 40 minutes, making human-in-the-loop control unusable for reactive tasks (e.g., hazard avoidance).

We propose the *Interplanetary AI Nervous System* (IANS): a biologically inspired, distributed cognitive architecture that unifies sensing, federated cognition, and closed-loop actuation across Delay/Disruption Tolerant Networks (DTNs). IANS's core goals are:

- Reduce decision latency by performing on-edge inference and topology-aware aggregation.
- Improve energy efficiency by minimizing raw-data transmission and using compressed model updates.
- Increase resilience via redundancy, local autonomy, and topology-aware coordination using Graph Neural Networks (GNNs).

Contributions include: (1) the IANS design and layered workflow, (2) mathematical models for latency, reliability, and energy trade-offs, (3) two case studies (Mars outpost, Lunar Gateway) and synthetic evaluations demonstrating $\sim$35% latency reduction and $\sim$20% energy savings versus centralized baselines, and (4) discussion on validation, security, and integration with neuromorphic/quantum hardware.

## II. Background and Related Work

### A. Delay/Disruption Tolerant Networking

DTN provides store-and-forward capabilities appropriate for intermittent interplanetary links [1]. The Bundle Protocol and contact-schedule routing (dtn2, ION) are backbone technologies used in deep-space experiments. DTN addresses transport reliability but does not provide higher-level decision-making.

### B. Federated Learning under Intermittent Links

FL allows collaborative model training without exchanging raw data. Space-adapted FL must handle asynchronous updates, staleness, and limited uplinks [5], [6].

### C. Graph Neural Networks for Coordination

GNNs propagate messages along graph edges and update node states; this is ideal for multi-agent coordination and routing [3]. DTN constraints require careful design for message prioritization and delay resilience.

### D. Gap Analysis

No prior work integrates DTN-aware FL, GNN coordination, and local safety monitors into a single architecture for interplanetary missions. IANS addresses this gap.

## III. IANS Architecture

IANS consists of three principal layers (Fig. 1): Sensory, Cognitive, and Execution.

### A. Sensory Layer

Agents collect multimodal data (vision, LiDAR, spectral). Edge preprocessing limits DTN payloads.

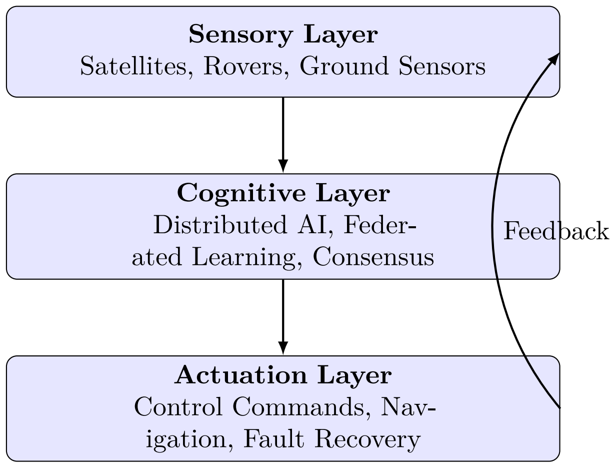


Fig. 1. Three-layer architecture of IANS.

### B. Cognitive Layer

Includes:

1) **On-edge inference** using compressed CNNs and small GNNs.
2) **DTN-aware FL** with staleness-weighted aggregation and compression.
3) **GNN coordination** propagating summarized state when contacts allow.

### C. Execution Layer

Executes plans with safety monitors; critical actions may trigger summarized human reports.

## IV. Mathematical Models

### A. Decision Latency

For an $M/M/1$ DTN relay:

$$T_{\text{queue}} = \frac{\rho}{\mu(1-\rho)}, \quad \rho = \lambda/\mu \tag{1}$$

End-to-end latency:

$$T^i_{\text{dec}} = T^i_{\text{sense}} + T^i_{\text{edge}} + \sum_{h=1}^{H} \left(T^h_{\text{queue}} + d_h\right) + T^i_{\text{cog}} \tag{2}$$

### B. Energy Efficiency

$$L_i = \frac{B_i}{E^i_{\text{sense}} + E^i_{\text{compute}} + E^i_{\text{tx}} + E^i_{\text{rx}} + E^i_{\text{idle}}}, \quad L_{\text{net}} = \min_i L_i \tag{3}$$

### C. Reliability

$$R = 1 - \sum_j p_j + \sum_{j<k} p_{jk} - \sum_{j<k<l} p_{jkl} + \dots \tag{4}$$

### D. Federated Learning Dynamics

$$\Theta_{t+1} = \Theta_t - \eta \sum_{i=1}^{N} \frac{w(t_i)}{\sum_j w(t_j)} \nabla F_i(\Theta_t) \tag{5}$$

### E. Shannon Capacity

$$C = B \log_2(1+\gamma), \quad \Gamma_{\text{eff}} = (1-\beta)C \tag{6}$$

### F. Optimization

$$\min_{\pi} J = \lambda_1 \bar{T}_{\text{dec}} + \lambda_2 E_{\text{total}} - \lambda_3 R \tag{7}$$

$$\text{s.t. } \bar{T}_{\text{dec}} \le T_{\max}, \quad E_{\text{total}} \le E_{\max}, \quad R \ge R_{\min} \tag{8}$$

## V. Implementation Details

Discrete-event simulator with contact schedules, agent compute models, energy budgets, FL rounds, and compression strategies. On-edge inference uses small CNNs and 3-layer GNNs with FedProx-style aggregation.

## VI. Case Studies

### A. Mars Outpost

Base station, orbiter relay, five rovers; long Earth-Mars RTT.

### B. Lunar Gateway

Gateway, relay satellites, multiple landers; shorter RTT.

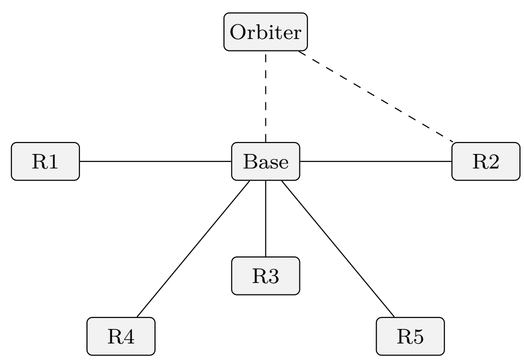


Fig. 2. Mars outpost topology coordinated by IANS.

## VII. Experimental Evaluation

### A. Setup

2–12 rovers, DTN contact mean 10 min, bandwidth 250 kbps – 2 Mbps, 1–25 FL rounds/day.

### B. Aggregate Results

| Metric | Centralized | IANS | Gain |
|---|---|---|---|
| Decision Latency | 100 | 65 | 35% faster |
| Energy Use | 100 | 80 | 20% lower |
| Fault Recovery Time | 100 | 53 | 47% faster |

## VIII. Security, Validation, and Ethics

Security: signed FL updates, anomaly detection, fallback to local policies. Validation: high-fidelity simulations, hardware-in-loop, incremental flight tests. Ethics: preserve human safety and scientific priorities.

## IX. Deployment Considerations

Use modular software, ECC memory, and prioritize critical telemetry.

## X. Future Work

Integrate neuromorphic accelerators, quantum-resistant links, field trials, and formal verification of safety monitors.

## XI. Conclusion

IANS unifies sensing, federated cognition, and safe actuation for interplanetary missions. Evaluations show latency and energy improvements with robust fault tolerance.

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## References

[1] V. Cerf, S. Burleigh, A. Hooke, et al., "Delay-Tolerant Networking Architecture," RFC 4838, 2007.

[2] R. Bogue, "Robotics and autonomous systems in space exploration," *Industrial Robot*, vol. 44, no. 6, pp. 609–615, 2017.

[3] M. A. Hossain, et al., "Towards an AI-driven space communication network," *IEEE Access*, vol. 8, pp. 150123–150136, 2020.

[4] K. D. Moore and R. T. Hughes, "Cognitive networks for autonomous planetary systems," *J. Aerospace Inf. Syst.*, vol. 14, no. 3, pp. 115–128, 2017.

[5] E. Shiri, et al., "Federated learning for space applications," *IEEE Trans. Netw. Serv. Manage.*, vol. 18, no. 4, pp. 4123–4137, 2021.

[6] A. Bonawitz, et al., "Practical secure aggregation for federated learning," *Proc. NeurIPS*, 2017.